

Steps to copy plan from one SQL id to another SQL id

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Context

This document explains steps to copy plan from one sql id to another sql id. This would be required in cases for eg hibernate release where sql id gets changed from old hibernate version to new version, having same sql text(functionally).

Get bad plan sql details

```
select sql_id, plan_hash_value, child_number, executions from gv$sql where sql_id='6b3ngypp9a0pz';
```

SQL_ID	PLAN_HASH_VALUE	CHILD_NUMBER	EXECUTIONS
6b3ngypp9a0pz	3265281732	0	6

Get good plan sql details

```
select sql_id, plan_hash_value, child_number, executions from gv$sql  
where sql_text not like '%v$sql%'  
and sql_fulltext like '%/*+ cardinality(t 10) */%';
```

SQL_ID	PLAN_HASH_VALUE	CHILD_NUMBER	EXECUTIONS
39xmr0csm20st	1957842980	0	1

OR

```
select sql_id, plan_hash_value, child_number, executions from gv$sql where sql_id='39xmr0csm20st';
```

Copy the plan across sql_id's

Note: Please use this method ONLY after 100% verification of bad and good SQL queries are exactly matching. Failing it may also have adverse impact on database stability.

```

set serverout on
declare
  v_sql_fulltext clob;
begin
  select sql_fulltext
  into v_sql_fulltext
  from v$sql
  where sql_id = '&sql_id_badplan'
  and child_number = &child_number_badplan ;
  dbms_output.put_line(v_sql_fulltext);
  dbms_output.put_line(
    dbms_spm.load_plans_from_cursor_cache(
      sql_id          => '&sql_id_goodplan',
      plan_hash_value => &plan_hash_goodplan,
      sql_text        => v_sql_fulltext,
      fixed           => 'YES',
      enabled         => 'YES'
    )
  );
end;
/
SQL>  2      3      4      5      6      7      8      9     10     11     12     13     14     15     16     17     18     19     20
Enter value for sql_id_badplan: 6b3ngypp9a0pz
old  7:      where sql_id = '&sql_id_badplan'
new  7:      where sql_id = '6b3ngypp9a0pz'
Enter value for child_number_badplan: 0
old  8:      and child_number = &child_number_badplan ;
new  8:      and child_number = 0 ;
Enter value for sql_id_goodplan: 39xmr0csm20st
old 12:      sql_id          => '&sql_id_goodplan',
new 12:      sql_id          => '39xmr0csm20st',
Enter value for plan_hash_goodplan: 1957842980
old 13:      plan_hash_value => &plan_hash_goodplan,
new 13:      plan_hash_value => 1957842980,

/* BeginningBalances.generate com.intuit.qbo.reports.QZReport */ select
sum(TxDetails.amount), Accounts.account_type_id, Accounts.account_type_id,
Accounts.number_order_by_name, Accounts.number_full_name from TxDetails_1
TxDetails, Accounts_1 Accounts, TxHeaders_1 TxHeaders where ((
TxHeaders.account_id IN ( (select /*+ cardinality(t 1) */ COLUMN_VALUE from
table(qbo_utils.str2tbl(:1 )) t) )) and TxDetails.tx_date < :2 and (
TxHeaders.tx_type_id IN ( (select /*+ cardinality(t 1) */ COLUMN_VALUE from
table(qbo_utils.str2tbl(:3 )) t) )) and TxDetails.is_no_post IS NULL and (
TxDetails.account_id IN ( (select /*+ cardinality(t 1) */ COLUMN_VALUE from
table(qbo_utils.str2tbl(:4 )) t) )) AND (Accounts.account_type_id NOT IN
(10,13,12,14,11) OR (TxDetails.tx_date >= :5 ))) and
TxDetails.account_id=Accounts.account_id(+) and TxDetails.tx_id=TxHeaders.tx_id
group by Accounts.account_type_id, Accounts.number_order_by_name,
Accounts.number_full_name order by Accounts.account_type_id,
Accounts.number_order_by_name, Accounts.number_full_name
1

PL/SQL procedure successfully completed.

```

Verify SQL plan baseline

```
set pages 0
set lines 1000
set head off
select * from dba_sql_plan_baselines where trunc(CREATED) = trunc(sysdate);

8.7829E+18 SQL_79e317155aa54fa8 /* BeginningBalances.
generate com.intuit.qbo.reports.QZReport */ select sum(TxD
SQL_PLAN_7mssr2pdaamx82073cfca KPOPAT

MANUAL-LOAD QBO_DATA

12.1.0.2.0 24-JUL-17 11.30.43.000000 PM 24-JUL-
17 11.30.43.000000 PM YES YES YES YES YES NO 128 SQL*Plus
1 47961 48000 112 0 0 0 0
```

Example

Get good plan sql details

```
SQL> select sql_id, plan_hash_value, child_number, executions from gv$sql where sql_id='1huyw2c9nsb8t';

SQL_ID          PLAN_HASH_VALUE CHILD_NUMBER EXECUTIONS
-----
1huyw2c9nsb8t    4239432339          0           3

1 row selected.

I SQL_PROFILE          SQL_PLAN_BASELINE          SQL_ID          CHILD_NUMBER
PLAN_HASH_VALUE EXECUTIONS EL_PER_EXEC TO_CHAR(LAST_ACTIVE_T LIO          PIO AVG_CPU_TIME_PER_EXEC
IO_TIME  AVG_ROWS FIRST_LOAD_TIME
- - - - -
Y
4239432339          3 .581703333 12-NOV 10:01          4207          1huyw2c9nsb8t          0
5377          1 2022-11-12/09:12:01          1038          .0567          .
```

Get bad plan sql details

```
I SQL_PROFILE          SQL_PLAN_BASELINE          SQL_ID          CHILD_NUMBER
PLAN_HASH_VALUE EXECUTIONS EL_PER_EXEC TO_CHAR(LAST_ACTIVE_T LIO          PIO AVG_CPU_TIME_PER_EXEC
IO_TIME  AVG_ROWS FIRST_LOAD_TIME
- - - - -
Y
1555738801          26 .326754231 12-NOV 10:09          5488          bzrmzug3h2ak6          0
305          1 2022-11-09/04:44:45          894          .0316          .
```

Copy the plan across sql_id's

```

SQL> set serverout on
declare
  v_sql_fulltext clob;
SQL> 2 3 begin
  select sql_fulltext
  into v_sql_fulltext
4 5 6 from v$sql
  where sql_id = '&sql_id_badplan'
7 8 and child_number = &child_number_badplan ;
  dbms_output.put_line(v_sql_fulltext);
  dbms_output.put_line(
9 10 11 dbms_spm.load_plans_from_cursor_cache(
12 sql_id => '&sql_id_goodplan',
13 plan_hash_value => &plan_hash_goodplan,
  sql_text => v_sql_fulltext,
  fixed => 'YES',
  enabled => 'YES'
  )
  );
end;
/ 14 15 16 17 18 19 20
Enter value for sql_id_badplan: bzrmzug3h2ak6
old 7: where sql_id = '&sql_id_badplan'
new 7: where sql_id = 'bzrmzug3h2ak6'
Enter value for child_number_badplan: 0
old 8: and child_number = &child_number_badplan ;
new 8: and child_number = 0 ;
Enter value for sql_id_goodplan: 1huyw2c9nsb8t
old 12: sql_id => '&sql_id_goodplan',
new 12: sql_id => '1huyw2c9nsb8t',
Enter value for plan_hash_goodplan: 4239432339
old 13: plan_hash_value => &plan_hash_goodplan,
new 13: plan_hash_value => 4239432339,
/* criteria query */ select count(distinct t0x1_.PAYROLL_RUN_SEQ) as y0_ from PSP_PAYCHECK this_ inner join
PSP_QBDT_PAYCHECK_INFO t1x2_ on this_.PAYCHECK_SEQ=t1x2_.PAYCHECK_FK inner join PSP_PAYROLL_RUN t0x1_ on this_.
PAYROLL_RUN_FK=t0x1_.PAYROLL_RUN_SEQ where (((this_.COMPANY_FK=:1 and t1x2_.COMPANY_FK=:2 ) and not (t0x1_.
PAYROLL_RUN_STATUS in (:3 ))) and not (lower(this_.SOURCE_PAYCHECK_ID) like :4 )) and ((not (t0x1_.
PAYROLL_RUN_TYPE in (:5 )) or (t0x1_.PAYROLL_RUN_TYPE in (:6 ) and t0x1_.PAYROLL_DIRECT_DEPOSIT_AMOUNT>:7 )) or
exists (/* criteria query */ select d1_.FINANCIAL_TRANSACTION_SEQ as y0_ from PSP_FINANCIAL_TRANSACTION d1_
inner join PSP_TRANSACTION_TYPE t0x1_ on d1_.TRANSACTION_TYPE_FK=t0x1_.TRANSACTION_TYPE_CD inner join
PSP_TRANSACTION_STATE t1x2_ on d1_.CURRENT_TRANSACTION_STATE_FK=t1x2_.TRANSACTION_STATE_CD where (((t0x1_.
ASSOCIATION_TYPE=:8 or t0x1_.TRANSACTION_TYPE_CD in (:9 , :10 )) and not (t1x2_.TRANSACTION_STATE_CD in (:11 ,
:12 ))) and d1_.SETTLEMENT_TYPE_CD=:13 ) and d1_.FINANCIAL_TRANSACTION_AMOUNT>:14 ) and d1_.
PAYROLL_RUN_FK=this_.PAYROLL_RUN_FK)))
1

PL/SQL procedure successfully completed.

```

Verify SQL plan baseline

```

SQL> set pages 0
set lines 1000
SQL> SQL> set head off
SQL> select * from dba_sql_plan_baselines where trunc(CREATED) = trunc(sysdate);
13067270237186397553 SQL_b558450c1f879571 /* criteria query */ select count(distinct t0x1_.
PAYROLL_RUN_SEQ) as y0_ from PS
SQL_PLAN_baq251hgsg5bj7a4a2317
INTUADMIN
MANUAL-LOAD-FROM-CURSOR-CACHE PSP_PRL_APP

```

```

19.0.0.0.0 12-NOV-22 10.16.43.440138
AM 12-NOV-22 10.17.00.000000
AM 12-NOV-22 10.17.00.000000 AM YES YES YES
YES YES NO 411 CompanyAdapter:435
getPayrollRunCount
3 1745110 169976
12622 3115 0 3

```

1 row selected.

```

SQL> @spb
13067270237186397553 SQL_b558450c1f879571 /* criteria query */ select count(distinct t0x1_.
PAYROLL_RUN_SEQ) as y0_ from PS
SQL_PLAN_baq251hgsg5bj7a4a2317
INTUADMIN
MANUAL-LOAD-FROM-CURSOR-CACHE
PSP_PRL_APP

```

```

19.0.0.0.0 12-NOV-22 10.16.43.440138
AM 12-NOV-22 10.17.00.000000
AM 12-NOV-22 10.17.00.000000 AM YES YES YES
YES YES NO 411 CompanyAdapter:
435
getPayrollRunCount
3 1745110 169976
12622 3115 0
3 3

```

1 row selected.

--After copy plan from good sql/old sql

I	SQL_PROFILE	SQL_PLAN_BASELINE	SQL_ID	CHILD_NUMBER
PLAN_HASH_VALUE	EXECUTIONS	EL_PER_EXEC	TO_CHAR(LAST_ACTIVE_T	PIO
IO_TIME	AVG_ROWS	FIRST_LOAD_TIME	AVG_CPU_TIME_PER_EXEC	
Y		SQL_PLAN_baq251hgsg5bj7a4a2317	bzrmzug3h2ak6	
1	4239432339	3 .062374 12-NOV 10:17	8833 26	.0497
.0127	1	2022-11-09/04:44:45		

1 row selected.